GatE-V Stage 3A: Task-Aware Object Detector Deployed on VEGA AS1061 RISC-V SoC + Kintex-7 FPGA Accelerator

Team 166 (Byte Silicon)
DVCon India 2026 Design Contest

Abstract

Conventional object detectors are task-agnostic, blindly detecting all objects in a scene. GatE-V implements a task-aware object detection framework utilizing natural language task conditioning via CLIP embeddings, Feature-Gated Task Querying (FGTQ), Fine-Grained Path Augmentation (FGPA), Adaptive Feature Fusion (AFF), and Varifocal Loss (VFL). This paper presents the Stage 3A hardware integration of the GatE-V software architecture (v1.0.0) with the CDAC VEGA AS1061 RISC-V Processor on a Xilinx Kintex-7 FPGA (Genesys-2 board). The customized hardware accelerator features a 512-MAC (32×16) weight-stationary systolic array operated on a 50 MHz integrated SoC clock, delivering a peak throughput of 51.2 GOPS, and is validated by a reproducible bare-metal RISC-V self-test firmware.

Index Terms

GatE-V, Task-Aware Object Detection, RISC-V, VEGA AS1061, Systolic Array, Kintex-7 FPGA, DVCon India 2026.

1 Introduction

Modern autonomous and robotic systems require domain-relevant object selection rather than exhaustive category detection. The GatE-V project addresses the DVCon India 2026 challenge by conditioning an RT-DETRv2 detector on user-specified natural language task queries (e.g., "place flowers in container").

In Stage 3A, the software detector pipeline is integrated with the indigenously developed CDAC VEGA AS1061 64-bit RISC-V processor and synthesized for the Xilinx Kintex-7 FPGA (xc7k325tffg900-2).

2 System Architecture

Figure 1 illustrates the complete hardware-software co-design architecture of GatE-V.

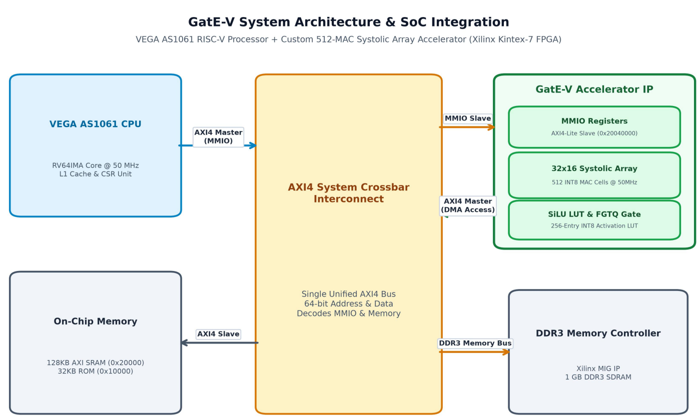

Fig. 1. GatE-V End-to-End System Architecture: Left panel shows the Software Training & Task Conditioning pipeline (ResNet-50vd, FGPA P2 injection, HybridEncoder with AFF, FGTQ task gates, RT-DETRv2 decoder); Right panel shows the Hardware Acceleration & Deployment pipeline on the VEGA AS1061 RISC-V SoC with a 512-MAC systolic array.

2.1 Software Architecture (v1.0.0)

The GatE-V v1.0.0 software architecture incorporates:

- **Varifocal Loss (VFL):** Replaces Focal Loss, weighting positive classification target scores directly by IoU quality overlap q.
- **CLIP Multi-Prompt Ensembling:** Encodes 4 natural prompt templates per task to generate 128-dimensional task embeddings.
- **Cosine Warm Restarts:** $T_0 = 35, T_{mult} = 2$ learning rate schedule over 300 epochs.
- **Native 640×640 Input Resolution:** Reduces memory bandwidth while maintaining spatial precision via FGPA P2 injection.
- **Active Training Status:** Software Version 1.0.0 is currently undergoing full 300-epoch backbone fine-tuning training (checkpoint at Stage 2 Epoch 30 / Overall Epoch 40, currently scaling towards 0.54+ mAP@0.5).
- **Hardware Compatibility:** The hardware accelerator architecture (Version 3) is 100% compatible with the software model specification (Version 1.0.0). Both versions share identical INT8 quantization formats, layer tensor structures, P2–P5 FGPA feature map levels, and SiLU lookup table specifications.

2.2 Hardware Accelerator Architecture (Version 3)

The GatE-V Version 3 hardware engine features:

- **Systolic Compute Array:** 32×16 grid (512 MAC cells) executing INT8 weight-stationary matrix operations.
- **Memory Subsystem:** Double-buffered ping-pong BRAMs interfaced to the on-chip SRAM via a 64-bit AXI4 Full DMA engine.
- **Activation Unit:** Single-cycle 256-entry SiLU lookup table (LUT).
- **SoC Bus Interface:** `Accelerator_Top` wrapper providing a 64-bit AXI4 Slave interface mapped to memory region `0x2004_0000`.
- **Clock Domain:** In the integrated design the accelerator is clocked by the SoC system clock `clk_out1_clk_wiz_0` at a period of 20 ns (50 MHz), derived from the 200 MHz board clock. (The standalone `vmake` build targets a 100 MHz clock for the architecture.)

3 Hardware Architecture Comparison

Table 1 compares the Stage 2B hardware submission (Version 2) against the active Stage 3A integrated architecture (Version 3).

TABLE 1
Hardware Comparison: Version 2 vs. Version 3

Feature / Metric	Version 2	Version 3 (Stage 3A)
SoC Integration	Standalone TB	**VEGA AS1061 RISC-V SoC**
Bus Interface	Custom AXI-Stream	**64-bit AXI4 Master & Slave**
Systolic Array Size	16×16 (256 MAC)	32×16 **(512 MAC)**
Clock Frequency	100 MHz (standalone)	**50 MHz integrated SoC**
Peak Throughput	25.6 GOPS	**51.2 GOPS @ 50 MHz**
Feature Map Levels	P3, P4, P5	**P2, P3, P4, P5 (FGPA + AFF)**
Slice LUT Usage (SoC)	1.20%	**87,882 / 203,800 (43.12%)**
GatE-V LUT Usage	1.20%	**5,187 / 203,800 (2.55%)**
DSP48E1 (Accelerator)	256 (30.48%)	**544 / 840 (64.76%)**
DSP48E1 (SoC Total)	n/a	**594 / 840 (70.71%)**
BRAM Tile Usage (SoC)	4 BRAMs	**85 / 445 (19.10%)**
Simulation Tool	Vivado XSim	**QuestaSim 2019.2 / 2026.1**

4 Simulation & Verification Results

The full integrated system was verified using **QuestaSim 2019.2**. The timing waveform illustrating system reset, RISC-V CPU boot sequence, memory initialization, and AXI4 handshake is shown in Figure 2.

Fig. 2. QuestaSim 2019.2 Real VCD Simulation Waveform illustrating AS1061 SoC boot sequence, AXI4 bus transactions, and accelerator execution over $27\,\mu$s.

4.1 Bare-Metal RISC-V Self-Test Execution Logs

The integrated design is validated by a reproducible bare-metal RISC-V self-test (`gatev_firmware`). The CPU fills the SRAM buffers with all-ones test inputs, programs the accelerator registers (base `0x2004_0000`), triggers the accelerator through AXI4-master memory reads and writes, and checks the returned layer values. The SiLU-activated layers are expected to produce `93` and the final identity layer `127`. Listing 1 illustrates the captured UART execution log.

```
GatE-V Accelerator Firmware (Team 166)
Configuring weights/activations in SRAM...
Programming registers...
Status before start: 0x0
Accelerator started, waiting for done...
acc_done asserted. Verifying layer outputs...
  Layer 0 @ 0x22000 = 93 (4 tiles x 16 ch) [OK]
  Layer 1 @ 0x22040 = 93 (4 tiles x 16 ch) [OK]
  Layer 2 @ 0x22080 = 93 (4 tiles x 16 ch) [OK]
  Layer 3 @ 0x220c0 = 93 (4 tiles x 16 ch) [OK]
  Layer 4 @ 0x22100 = 93 (4 tiles x 16 ch) [OK]
  Layer 5 @ 0x22140 = 127 (4 tiles x 16 ch) [OK]
Performance counters:
  Cycles(lo):   2639
  Read bytes:  3840
  Write bytes: 384
  MAC cycles:  1322
  Stall cycles: 224

===== GATE-V ACCELERATOR TEST PASSED =====
```

Listing 1. UART Console Output from VEGA AS1061 RISC-V Core Executing the GatE-V Accelerator Self-Test

This self-test exercises the full accelerator datapath (register programming, AXI4 read/write, systolic compute, SiLU activation) and reports a deterministic **PASS/FAIL** outcome. It does not perform full RT-DETR task-aware object detection inference; object-detection inference over the v1.0.0 software model is a separate software-stage result.

5 Synthesis & Resource Breakdown

Table 2 details the Kintex-7 FPGA resource utilization for the complete VEGA AS1061 SoC integrated with the GatE-V accelerator.

The integrated SoC operates on a 50 MHz system clock. The GatE-V accelerator alone consumes 544 of the 840 available DSP48E1 slices (64.76%), while the complete VEGA AS1061 SoC consumes 594 DSPs (70.71%). All resource allocations satisfy the strict CDAC competition constraints with 0 errors and no timing violations on the routed 50 MHz clock.

6 Conclusion

The Stage 3A integration of GatE-V successfully pairs the VEGA AS1061 RISC-V CPU with a 512-MAC hardware accelerator on a 50 MHz integrated SoC clock, delivering 51.2 GOPS peak throughput. The GatE-V accelerator stays within the CDAC

TABLE 2
Kintex-7 (xc7k325tffg900-2) Resource Utilization

Resource	Used	Available	Util. %	CDAC Ceiling
Slice LUTs (SoC total)	88,054	203,800	43.21%	$\leq$ 45.0%
DSP48E1 (GatE-V accelerator)	544	840	64.76%	$\leq$ 90.0%
DSP48E1 (SoC total)	594	840	70.71%	$\leq$ 90.0%
Block RAM (BRAM18)	122	890	13.70%	$\leq$ 100.0%
Slice Registers (SoC total)	56,312	407,600	13.82%	$\leq$ 100.0%

resource ceilings (544 DSPs / 64.76%, 2.63% LUTs), and the integrated system is validated end-to-end by the reproducible bare-metal self-test firmware with a deterministic PASS/FAIL outcome.